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SHENZHEN FUMAN ELECTRONICS CO., LTD

FM6363 Control and Programming Guide V0.1

FM6363 Control and Programming Guide

One control command

The chip contains a simple 16-bit shift register into which grayscale values and configuration values are latched.

Control commands are parsed by counting the length of the LE signal; different LE lengths represent different commands. For example, a length of 1...

The LE signal represents the "Data Latch" command, used to control the shift register to latch grayscale values, changing the value in the shift register...

16 bits of data are sent to SRAM. Table 1 lists all commands and their meanings.

Table 1. Control Commands:

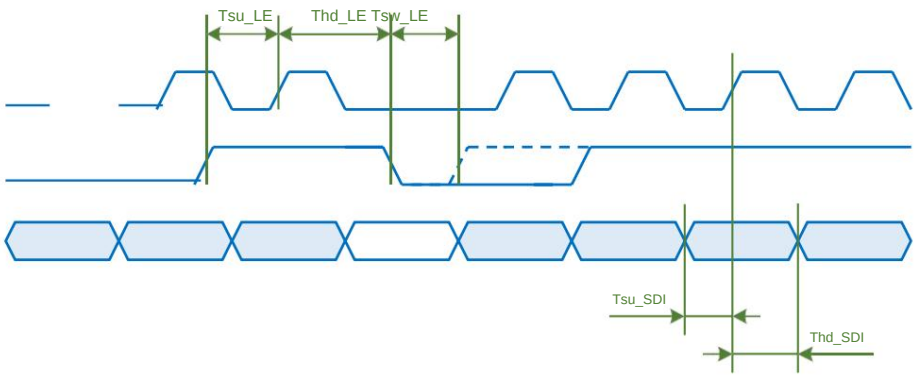
Command Name	**LE**	Instruction Description
DATA_LATCH	1. Latch 16 bits of data and send them to SRAM	
WR_DBG	2. Write to debug register 5.	
VSYNC	3. Update display data	
WR_CFG1	4. Write configuration register 1	
WR_CFG2	6. Write configuration register 2	
WR_CFG3	8. Write configuration register 3	
WR_CFG4	10 Write Configuration Register 4	
EN_OP	12 Enable all output channels	
DIS_OP	13. Close all output channels	
PRE_ACT	14. Write enable	
MBIST	15 Enable SRAM checksum and read status	

Note 1: The length of LE refers to the number of rising edges of DCLK when LE is high. As shown in Figure 2, the first...

The length of the LE signal is 1, which means the command is the "Data Latch" command.

Note 2: PRE_ACT and EN_OP are sent first for each frame, and then the registers are configured. Before configuring each register, PRE_ACT and EN_OP must also be sent first.

PRE_ACT, LE waveform



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The setup and hold times for the LE and SDI signals are shown in the table below.

Signal name	MIN	Remark
Tsu_LE	7ns	
Thd_LE	7ns	
Tsw_LE	10ns	
Tsu_SDI	3ns	
Thd_SDI,	3ns	

The grayscale data is latched using the "Data Latch" command, with the first 16-bit data serving as the first row number for channel 15. According to the data, the second 16-bit data is used as the first row of data in channel 14, and the 17th 16-bit data is used as the first row of data in channel 15. Two rows of data. Table 3 shows the corresponding data instructions for all 64 rows of data.

Data Order	Line	Channel
1	Line 1	Channel 15 (OUT15)
2		Channel 14 (OUT14)
■ ■ ■		■ ■ ■
16		Channel 0 (OUT0)
17	Line 2	Channel 15 (OUT15)
18		Channel 14 (OUT14)
■ ■ ■		■ ■ ■
32		Channel 0 (OUT0)
... ...		
497	Line 64	Channel 15 (OUT15)
498		Channel 14 (OUT14)
■ ■ ■		■ ■ ■
512		Channel 0 (OUT0)

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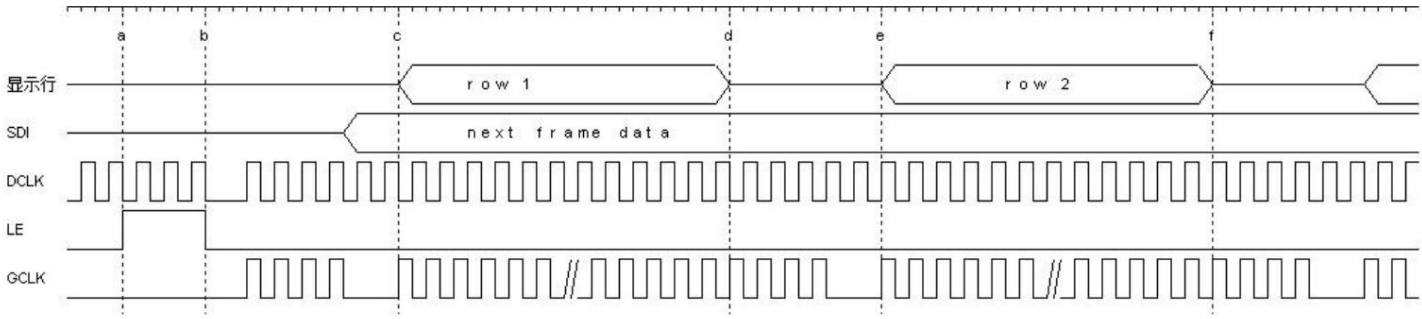
Display timing

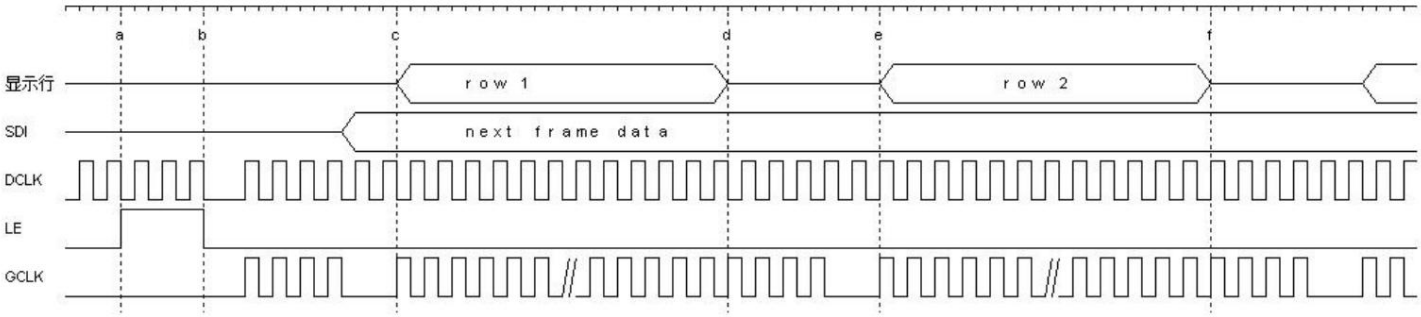


Display timing

The timing sequence is shown in the diagram above:

a) From time a to time b, first send VSYNC (LE high level for 3 DCLK widths) to start a frame change. After time b, you can start latching the data for the next frame; b) From time b to time c, send 4 GCLKs; c) From time c to time e, the display time for one line, a total of 74 GCLKs; d) Time d, line feed time; e) From time d to time e, line feed time. After time d, you need to send 4 more GCLKs (included in...) Within 74GCLK); f) At time e, the next line begins.

Configuration register timing



Configuration register timing

The timing diagram for configuring the registers is shown above:

g) From time a to time b, send PRE_ACT (LE high level of 14 DCLK width); h) From time c to time d, send EN_OP (LE high level of 12 DCLK width); i) From time e to time f, send PRE_ACT; j) From time g to time h, write register 1 (LE high level of 4 DCLK width); k) From time i to time j, send PRE_ACT; l) From time k to time l, write register 2 (LE high level of 6 DCLK width); m) Configure register 3 and register 4 according to steps k) and l).

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Two registers

Register 1

Bit	Name	Default	Description
15	OPEN_DET		Open circuit detection 0: Close 1: Open
14	GCLK_N		This bit controls the number of GCLKs in the frame and line. 1: The frame header contains 6 GCLKs, and each subsequent line contains 78 GCLKs; 0: The frame header contains 4 GCLKs, followed by 74 GCLKs per line;
13:8	SCAN_LINE		Number of lines scanned 0: Line 1 1: 2 lines 2: 3 lines ... Lines 63 and 64
7:6	OPT		Low gray speckle/high brightness coupling optimization 00: Test Mode 01: Level 2 optimization (recommended value) 10:3 level optimization (optimized highlight coupling recommended value) 11: Test Mode
5:4	TEST		CFG1<5:4>=01&CFG3<9:8>=11 Enable REG5<8:0> CFG1<5:4>=01 Enable REG4<3:1> CFG1<5:4>=01&CFG3<9:8>=01 Configuration for enabling CFG_DBGX (not listed)
3	Crossboard coupling		0: Close 1: Open
2:0	TEST		Force minimum grayscale output: 001~111 output grayscale values 1~7 respectively.

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Register 2

Bit	Name	Default	Description
15	TEST		0: Row scan configuration 33~64 scans (default) 1: Line scan configuration 1~64 scans
14:10	ADJ		Blanking control register Levels 1-31 correspond to registers respectively. cfg2[14:10]=00000-11111 Enable register cfg3[2] Recommended R=31, G=28, B=23
9	I_DIV4N		Current range adjustment 1: High current setting 0: Low current setting
8:1	IGAIN		Constant current source output configuration register IOUT=19*IGAIN/(Rext*256) @ I_DIV4N=1 IOUT=19*IGAIN/(Rext*1024)@ I_DIV4N=0 Igain ÿ 64 (Required)
0			Hidden mode 0: Fixed hidden line removal (default) 1: Pulse blanking

Register 3

Bit	Name	Default	Description
15	Reserved	--	
14:12	TEST		Adjust VDS (for core circuits), range 160mV~520mV The configured VDS levels increase sequentially from 000 to 111.
11:10	TEST		PWM output delay Default configuration "00"
9:8	TEST		scan chain configuration You must configure CFG1<5:4>=01 before you can write to it.
7:4	PWM _ALL		Low gray color cast compensation adjustment Levels 1-16 correspond to registers cfg3[7:4]=1111-0000 respectively. Enable register cfg4[14]
3	TEST		Frequency multiplier switch 0: Close 1: Open

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2	UP_SEL		Shadow removal switch: 1: Open 0: Off
1:0	VLDO		VLDO voltage 4 adjustable voltage levels

Register 4

Bit	Name	Default	Description
15	TEST		Hidden shadow reduction and widening by 1GCLK 0: Close 1: Open
14	PWM_ADD_EN		Low gray color cast compensation switch 1: Open 0: Close
13	TEST		SDI to SDO delay control 0: Delay 1: No delay
12	Non_clp_en		0: Close 1: Open
11:10	TEST		Channel closing speed 00: Slowest 11: Fastest
9:8	TEST		Channel opening speed 00: Slowest 11: Fastest
7	OPEN_SCAN		Open-circuit data write enable 0: No update 1: Updatable
6	TEST		Clamp switch 0: Close 1: Open
5:4	TEST		Between blanking (V+) and clamping (V-) The two highest two digits of the difference VA
3	TEST		Forced PWM output full brightness grayscale 0: Close 1: Open Used in conjunction with opening a circuit
2	TEST		Black screen energy-saving switch 0: Close 1: Open
1	VRG_BUF		VRG_BUF bias current 0: Default 1: Double

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0	TEST		The grayscale level is halved. 0: Off (default) 1: Open

Register 5

Bit	Name	Default	Description
15	Reserved	/	--
14			Clamping/Blanking Enhancement 0: Not enabled (default) 1: Enable
13	Non_clp_v_sel		0: (Default) 1:
11:12			Between blanking (V+) and clamping (V-) The lower two bits of the difference VA
10			Hidden beam enhancement 0: Not enabled (default) 1: Enable
9			Clamping enhancement 0: Not enabled (default) 1: Enable
8			CFG1~4 Value Reading 0: Close 1: Open
7	Reserved	/	--
5:6	TEST		00: SDO outputs SDI 01: SDO output is always 0 10: SDO outputs LDO 11: SDO output VGR
4:2			010 Automatic current range adjustment 000/001/111 correspond to the 4th gear in automatic transmission.
0:1			00/01 Open Circuit Detection 11: Forced road opening 10: Forced Normal

Main interface settings

1. Gray levels

The FM6363 uses a dual-edge display; please calculate the grayscale levels using the dual-edge **GCLK** method.

2. Refresh Rate

The refresh rate is fixed at 8x, with 74 GCLKs per line. The register values are cfg1[5:4]=11 and cfg1[14]=0.

3. Refresh rate

Fixed 8x multiplier, recommended refresh rate **3840Hz**

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Extended Interface

Suggestions 1.

Blanking Settings Enable the blanking function using the checkbox menu. When checked, cfg3[2]=1; when unchecked, cfg3[2]=0. Adjust blanking capability using the left and right sliders, and the right side with the up and down selection menu. There are a total of 32 levels, with levels 0-31 corresponding to registers cfg2[14:10]=00000-11111 respectively.

The Default value is checked, and the ratings are: R=31, G=25, B=23.

2. Low gray-white balance adjustment

Enable low grayscale balance adjustment using the checkbox menu. When checked, cfg4[14]=1; when unchecked, cfg4[14]=0. For low grayscale balance adjustment, use the left... The right slider, along with the up and down selection menus on the right, has a total of 16 levels. Levels 1-16 correspond to registers cfg3[7:4]=1111-0000 respectively.

The default value is unchecked.

3. The low-gray-dot effect is

optimized using an up-and-down selection menu with 4 levels. Levels 1-4 correspond to registers cfg1[7:6]=00-11 respectively . **The default** value is level **2.** It is recommended to use level 2 as the default.

4. LED Open Circuit Cross Elimination 1. Turn off GCLK. 2. Write reg1[15]=1, reg4[7]=1, send 5 GCLKs (to clear the original open state). 3. Send VSYNC command. 4. Send 16'hffff (all 1) data to all pixels.

5. Change the A signal in the row address ABCDE from 0 to 1, wait for 10 clks, and then change the A signal from 1 to 0 (to prevent 74LS74 from locking). 6. Send VSYNC command 7. Send 16 GCLKs 8. Wait 400us, send the remaining GCLKs, and add a newline 9. Repeat steps 7 to 8, send all the GCLKs of all rows, and complete the first set of data display for all rows 10. Write reg4[7]=0 to complete the open circuit detection

FM6363 Disable open circuit detection: Write reg1[15]=0, reg4[7]=0

5. Add a current

gain button, with the setting method the same as the external current gain and linked accordingly. The current gain corresponds to register cfg2[9:1]. Igain=DEC cfg2[8:1]. When cfg2[9]=1,

Iout=18.5x Igain/(Rext x 256

255ÿIgainÿ64 50%-200%

when cfg2[9]=0, Iout=18.5x Igain/(Rext x 1024

255ÿIgainÿ64 12.5-50%

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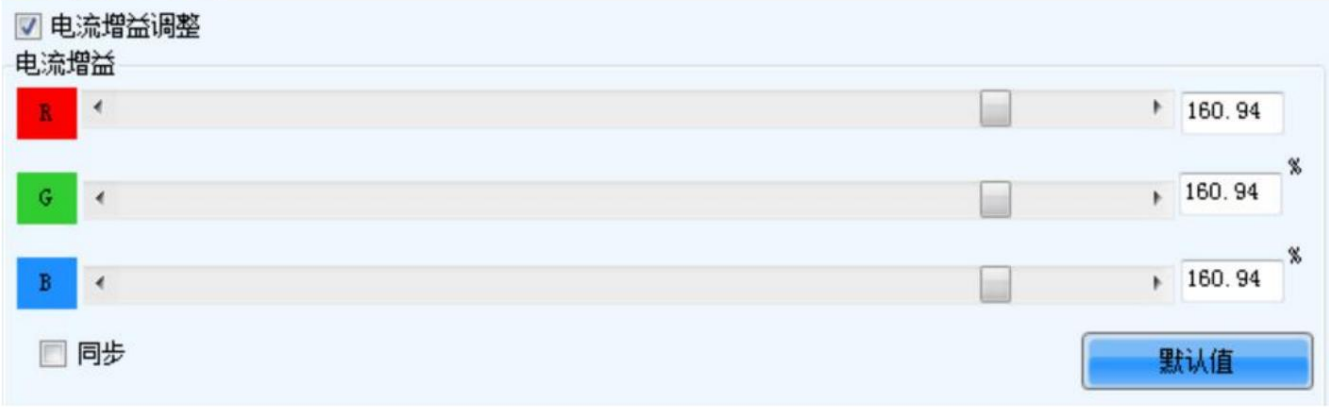

Extended advanced settings

1. Add an advanced settings button to the extended properties; clicking the button will bring up the advanced settings interface.

2. Add a 'Default Settings' button. Clicking this button will restore all register values to their default values. A confirmation menu should be added to prevent accidental changes. 3. Add a 'Last Modified' button to remember the last manually modified value. Clicking this button will restore the last modified value. 4. Add four sets of register input boxes (hexadecimal). 5. The register values on this interface should be synchronized with the extended properties interface.

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Line break timing suggestions

1. It is recommended to write to one set of registers per frame, completing the write in four frames, or configure the registers within the display time.

2. Minimize frame interval time. It is recommended to adjust the line break time appropriately to minimize the blank display time within a single frame. 3. Adjust the display timing: after sending VSYNC, send two GCLKs first, then proceed with normal display, and place a line break on the second-to-last GCLK of each line.

(Alternatively, the afterglow control signal can be sent two GCLKs in advance.)